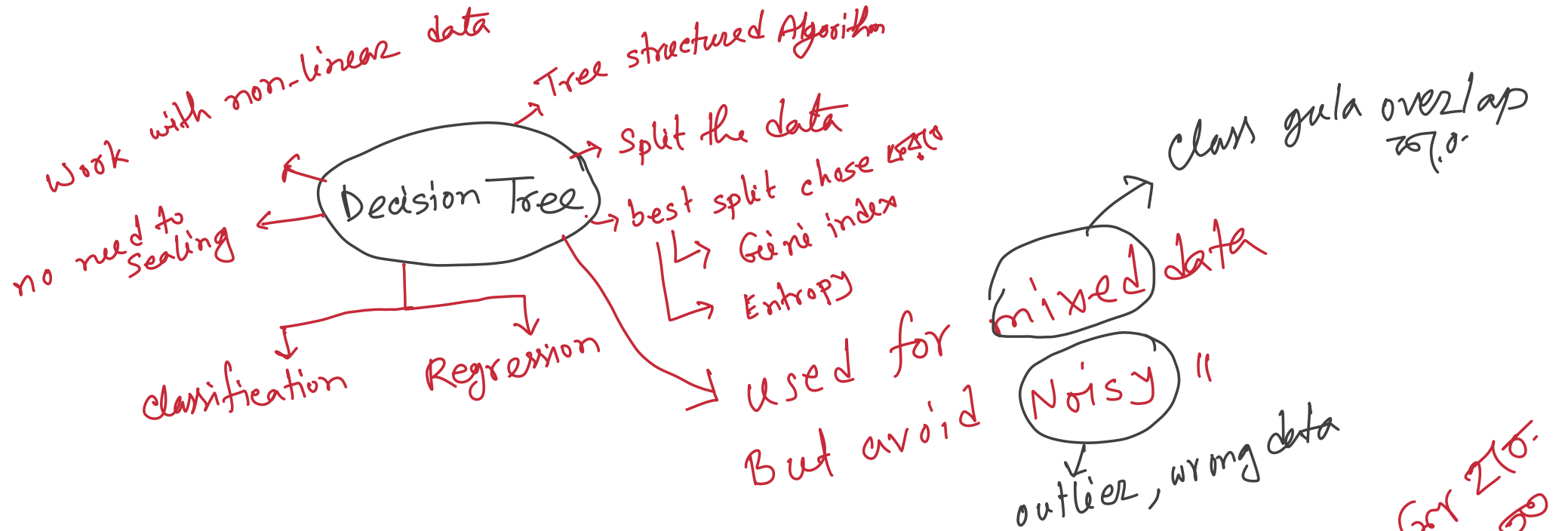




👉 "Should I play cricket today?"

- Is it raining?
 - Yes → Don't play
 - No → Go next
- Is it sunny?
 - Yes → Play
 - No → Maybe

👉 This is exactly a decision tree

Structure of Decision Tree

🔴 Visual idea



Decision tree diagram example

Decision tree

Decision tree

Parts:

- Root Node → starting point
- Decision Node → condition (e.g., marks > 50?)
- Leaf Node → final answer (Pass/Fail)
- Branches → yes/no paths

How it works (step-by-step)

- Take dataset
- Choose best feature (question)
- Split data
- Repeat for each branch
- Stop when:
 - Data pure OR
 - Depth limit reached

```
Python

from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

depth 250 (for 210)

data memorize

for 210



Model selection depends on data characteristics. For linear data we use Logistic Regression or Linear SVM. For non-linear structured data we use SVM with kernel or Decision Tree. For noisy data we prefer simpler models. For large datasets SGDClassifier is efficient, and for interpretability Decision Tree is preferred.